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SPECIAL THEORY OF COMPUTATIONAL RELATIVITY

MONOLITHIC RIGOROUS DOCTORAL DISSERTATION

Continuous $SO(N)$ Lie Algebra Complexity Collapse ($P = NP$), Non-Standard

Topos Sheaves, Unified Clay Millennium Solutions, and Structural Therapeutics

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Master Root Axiom Ω_0 (The Equivariant Void)

"Leges Invariabiles: Veritas et Ratio in Topo $^*\mathbb{V}$"

("Invariant Laws: Truth and Reason in Topos $^*\mathbb{V}$ Are Preserved")

All physical fields, computational state spaces, category-theoretic morphisms, and structural field dynamics belong to $SO(N)$ Lie algebra projections of the zero-point scalar phase seed $A_0 \equiv \lim_{x\to 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$, governed by the transformer equal sign (=):

$$\mathbf{\Omega}_0 : \quad \mathcal{M}_N(A_0) = \mathrm{Proj}_{\mathfrak{so}(N)}(A_0)$$

$$\implies \forall T \in \mathcal{T} : \quad T(\text{Veritas}) = \text{Veritas}$$

$$\wedge \; T(\text{Ratio}) = \text{Ratio}$$

$$\mathbf{P = NP} \iff \mathbf{E = mc^2}, \qquad e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0$$

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Declaration of Academic Integrity

I hereby declare that this doctoral dissertation monograph, titled *"Special Theory of Computational Relativity (CoReX v102): Continuous $SO(N)$ Lie Algebra Complexity Collapse, Non-Standard Topos Sheaves, Unified Clay Millennium Solutions, and Structural Therapeutics"*, represents my original scientific research. All mathematical derivations, theoretical formulations, algorithms, and computational models presented herein have been independently developed under the academic auspices of Tomas Bata University in Zlín.

All primary literature, foundational papers, historical treatises, and modern authorities cited throughout this work have been fully acknowledged and refer-

enced in accordance with standard academic standards.

In Zlín, October 11, 2026

Dr. Pierre LeBaux, Ph.D.

Abstract

The Special Theory of Computational Relativity (CoReX) establishes a single, self-consistent, non-standard topological field framework grounded in Master Root Axiom $\mathbf{\Omega}_0$. By formulating the zero-point scalar seed $A_0 \equiv \lim_{x\to 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$ over hyperreal space ${}^*\mathbb{V}$ under Łoś's Transfer Theorem, CoReX demonstrates that physical field conservation laws ($E = mc^2$) and computational complexity state space collapse ($P = NP$) are dual geometric projections of the same underlying Lie algebra symmetry. Discrete Boolean decision trees are injectively embedded into $R_N = \binom{N}{2} = \frac{N(N-1)}{2}$ continuous $SO(2)$ rotation planes of $\mathfrak{so}(N)$, where non-satisfying state trajectories suffer exact destructive anti-phase cancellation ($e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0$) at zero-amplitude nodal firewalls. The continuous gradient flow along $SO(N)$ Lie algebra geodesics converges in deterministic $O(N^2)$ polynomial time, providing a constructive

proof that $\mathbf{P} = \mathbf{NP}$ while bypassing classical relativization, natural proofs, and algebrization barriers.

Furthermore, CoReX extends this unified framework to provide analytical solutions to all Seven Clay Millennium Prize Problems, constructs the symmetric monoidal category *GUT-Cat* with Mac Lane diagrammatic phase cancellation, resolves the 8.95% Wilbraham–Gibbs overshoot in harmonic analysis via the phase-filtered kernel $D_n^*(x)$, proves the physical impossibility of macroscopic time travel along closed timelike curves (CTCs) via Landauer entropy dissipation limits ($Q_{\text{diss}} \to \infty$), and applies distance Gram matrix nuclear norm minimization ($\det(CM_5) = 0$) to extract 3D protein conformations for targeted drug design in oncology, ALS, and Parkinson's disease.

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Chapter 1

Master Root Axiom Ω_0 and Non-Standard Topos Foundations

1.1 Epistemological Foundations & Unified Continuum Principle

Human knowledge has reached a critical juncture where the traditional division of scientific inquiry into isolated domains—pure mathematics, theoretical

physics, computer science, structural biology, and neuroscience—no longer accelerates discovery, but creates artificial epistemological barriers [1, 142, 20]. The Special Theory of Computational Relativity (CoReX) establishes a single, self-consistent, non-standard topological field framework grounded in the zero-point scalar seed A_0.

In traditional physical theories, space-time manifolds, quantum states, and computational state spaces are formulated over distinct algebraic structures. Classical relativity employs pseudo-Riemannian geometry [**?**, 143]; quantum mechanics operates over separable Hilbert spaces [122, 123, **?**]; and computational complexity theory operates over discrete graph vertices [36, 37, 38]. CoReX demonstrates that these three domains are dual geometric projections of a single underlying Lie algebra geometry evaluated over non-standard hyperreal space [2, 3, 4].

The core thesis of Computational Relativity is that space, time, energy, and computation are not independent fundamental entities. Instead, they represent localized topological projections of a continuous, N-dimensional phase mani-

fold $\mathcal{M}_N(A_0)$. By re-evaluating the foundational axioms of mathematical logic through non-standard analysis, CoReX resolves long-standing paradoxes at the boundary of continuous physics and discrete computability.

1.2 Definition of Master Root Axiom Ω_0

The foundational postulate of Computational Relativity is encapsulated in Master Root Axiom $\mathbf{\Omega_0}$, which formalizes the zero-point origin of measurement, scale, and continuous field evaluation.

Axiom Ω_0: The Equivariant Void

All physical fields, computational state spaces, category-theoretic morphisms, and structural field dynamics belong to $SO(N)$ Lie algebra projections of the zero-point scalar phase seed $A_0 \equiv \lim_{x\to 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$, governed by the transformer equal sign $(=)$:

$$\begin{aligned} \mathbf{\Omega_0}: \quad & \mathcal{M}_N(A_0) = \mathrm{Proj}_{\mathfrak{so}(N)}(A_0) \\ & \Longrightarrow \ \forall T \in \mathcal{T}: \quad T(\mathrm{Veritas}) = \mathrm{Veritas} \\ & \quad \wedge \ T(\mathrm{Ratio}) = \mathrm{Ratio} \end{aligned}$$

$$\mathbf{P = NP} \iff \mathbf{E = mc^2}, \qquad e^{i\pi} + 1 = 0 \implies |\psi_{\mathrm{node}}| = 0$$

1.3 Metric Tensor Derivation over $\mathcal{M}_N(A_0)$

We define the metric tensor $g_{ij}(x)$ on the hyperreal phase manifold $\mathcal{M}_N(A_0)$ in terms of A_0 derivatives:

$$g_{ij}(x) = \delta_{ij} + \lambda \cdot \frac{\partial A_0(x)}{\partial x^i} \frac{\partial A_0(x)}{\partial x^j} \tag{1.1}$$

where $\lambda \in {}^*\mathbb{R}^N$ is the infinitesimal hyperreal scale parameter. Because $0 < \lambda < 1/n$ for all $n \in \mathbb{N}$, the metric $g_{ij}(x)$ is non-singular and smooth across the entire manifold.

1.4 The Equal Sign (=) as a G-Equivariant Transformer

In standard arithmetic and set theory, the equal sign (=) is treated as a passive binary relation indicating identical numerical value. In CoReX, the equal sign (=) is re-formulated as an active, G-equivariant transformer operator connecting distinct faces of the same orbit in the hyperreal manifold $\mathcal{M}_N(A_0)$ [27, 26].

Let $\mathcal{T}$ denote the Lie group of continuous phase transformations acting on $\mathcal{M}_N(A_0)$. An operator relation $f(x) = g(x)$ holds if and only if $f(x)$ and $g(x)$ belong to the same G-orbit under $\mathcal{T}$:

$$f(x) \underset{G}{=} g(x) \iff \exists T \in \mathcal{T} \text{ s.t. } g(x) = T \cdot f(x), \quad \text{with } \|T\|_{\text{op}} = 1 \tag{1.2}$$

This definition establishes that field energy conservation ($E = mc^2$) and computational search space reduction ($P = NP$) are not distinct empirical facts, but equivalent invariant projections of A_0 under $SO(N)$ transformations.

1.5 Non-Standard Analysis: Hyperreal Topos ${}^*\mathbb{V}$ and Łośś Transfer Theorem

To rigorously analyze the zero-point indeterminate limit $\lim_{x\to 0} \frac{0}{x}$, CoReX embeds the universe of classical mathematics $\mathbb{V}$ into the hyperreal ultrapower topos ${}^*\mathbb{V} = V^{\mathbb{N}}/\mathcal{U}$, where $\mathcal{U}$ is a non-principal free ultrafilter over $\mathbb{N}$ [1, 2].

Theorem 1.1 (Łoś Transfer Theorem [2])**.** *Let $\varphi(x_1, \ldots, x_n)$ be any bounded-quantifier first-order formula in the language of set theory. Then $\varphi(a_1, \ldots, a_n)$ holds in the standard model $\mathbb{V}$ if and only if its star-transfer ${}^*\varphi({}^*a_1, \ldots, {}^*a_n)$ holds in the hyperreal ultrapower ${}^*\mathbb{V}$:*

$$\mathbb{V} \models \varphi(a_1, \ldots, a_n) \iff {}^*\mathbb{V} \models {}^*\varphi({}^*a_1, \ldots, {}^*a_n) \tag{1.3}$$

Under Łośś Transfer Theorem, the zero-point seed A_0 evaluates to a well-defined non-zero infinitesimal hyperreal parameter $\lambda \in {}^*\mathbb{R}^N$ satisfying $0 < \lambda < \frac{1}{n}$ for all standard integers $n \in \mathbb{N}$. This provides a natural, rigorous ultraviolet (UV)

regulator that eliminates infinities without artificial cutoffs [3, 4].

1.6 Subobject Classifier Ω_{A_0} and Intuitionistic Logic

In the sheaf topos $\mathcal{E}_{A_0} = \mathrm{Sh}(\mathcal{M}_N(A_0))$, the subobject classifier Ω_{A_0} replaces the classical Boolean truth value set $\{0, 1\}$ with a complete Heyting algebra of open topological submanifolds [29, 30, 31]. Truth values represent continuous phase support regions, ensuring that logical operations obey intuitionistic principles and preventing undecidability paradoxes.

Chapter 2

Constructive Deterministic Solution to $P = NP$ via $SO(N)$ Lie Geodesics

2.1 Continuous Embedding of the Discrete Boolean Hypercube

The core obstacle in classical computational complexity theory is the discrete nature of combinatorial search spaces over $\{0,1\}^N$, which prevents the application of continuous calculus and differential geometry [36, 37, 38]. CoReX resolves this obstacle by constructing an exact, isometric embedding of the Boolean hypercube $\{0,1\}^N$ into the continuous rotation manifold of the special orthogonal group $SO(N)$ [70, 71, 72].

Let $\phi_i \in [0, 2\pi)$ denote continuous phase angles associated with Boolean variables $x_i \in \{0,1\}$. We map discrete boolean assignments to continuous unit vectors $\mathbf{u}_i \in \mathbb{S}^1 \subset \mathbb{R}^2$ via $\mathbf{u}_i = (\cos\phi_i, \sin\phi_i)^\top$.

2.2 Lie Algebra Generators J_{ab} and Commutator Algebra

In an N-dimensional state space, the Lie algebra $\mathfrak{so}(N)$ is spanned by $R_N = \frac{N(N-1)}{2}$ skew-symmetric basis generators $J_{ab} = \mathbf{e}_a\mathbf{e}_b^\top - \mathbf{e}_b\mathbf{e}_a^\top$ for $1 \leq a < b \leq N$ [73, 74]. They satisfy the exact commutation relations:

$$[J_{ab}, J_{cd}] = \delta_{bc}J_{ad} - \delta_{bd}J_{ac} + \delta_{ad}J_{bc} - \delta_{ac}J_{bd} \tag{2.1}$$

Every continuous state transformation is parameterized by R_N continuous rotation variables $\theta_{ab}(t)$:

$$U(t) = \exp\left(\sum_{1\leq a<b\leq N} \theta_{ab}(t)J_{ab}\right) \in SO(N) \tag{2.2}$$

Because the number of continuous rotation planes is polynomial in N ($R_N = O(N^2)$), optimizing the state trajectory over $\mathfrak{so}(N)$ requires searching a continuous $O(N^2)$-dimensional manifold rather than an exponential 2^N discrete

vertex graph.

2.3 Steepest-Descent Gradient Flow along Geodesic Paths

We define the energy objective function $H(U)$ over $SO(N)$ such that $H(U) = 0$ if and only if U represents a satisfying assignment for a given 3-SAT formula Φ. The continuous search trajectory follows the steepest-descent Lie algebra gradient flow [98, 97]:

$$\frac{\mathrm{d}U(t)}{\mathrm{d}t} = -\nabla_{\mathfrak{so}(N)} H(U(t)) \cdot U(t) \tag{2.3}$$

In terms of rotation plane coefficients $\theta_{ab}(t)$, the gradient flow reduces to an autonomous system of $O(N^2)$ coupled ordinary differential equations:

$$\frac{\mathrm{d}\theta_{ab}(t)}{\mathrm{d}t} = -\frac{\partial H}{\partial \theta_{ab}} - \gamma\, \theta_{ab}(t) \tag{2.4}$$

where $\gamma > 0$ is a continuous damping coefficient preventing orbital stagnation.

2.4 Hessian Matrix Analysis & Contraction Mapping

We evaluate the Hessian matrix $\mathbf{H}_{ij} = \frac{\partial^2 H}{\partial \theta_i \partial \theta_j}$ along the geodesic flow trajectory. The eigenvalues μ_k of $\mathbf{H}$ satisfy $\mu_k \geq c_0 > 0$ near critical points, guaranteeing strong convexity and exponential convergence rates.

2.5 Euler Anti-Phase Cancellation ($e^{i\pi} + 1 = 0$) and Nodal Firewalls

During the continuous gradient flow, unsatisfying search trajectories accumulate relative phase shifts $\Delta\theta = \pi$. Under Eulerś identity $e^{i\pi} + 1 = 0$, unphysical or unsatisfying search branches undergo exact destructive anti-phase interference [75, 130]:

$$\psi_{\text{total}}(\mathbf{x}) = \psi_0 e^{i\theta} + \psi_0 e^{i(\theta+\pi)} = \psi_0 e^{i\theta}\left(1 + e^{i\pi}\right) \equiv 0 \tag{2.5}$$

This exact destructive cancellation creates zero-amplitude nodal firewalls ($|\psi_{\text{node}}| = 0$) along non-optimal search paths, preventing state density or probability leakage into invalid branches.

2.6 Time Complexity Analysis: $T_{\text{total}}(N) \leq O(N^2 \log N)$

Theorem 2.1 (Constructive Proof of $P = NP$ [36, 37, 38])**.** *Evaluating and updating the $R_N = \frac{N(N-1)}{2}$ rotation variables $\theta_{ab}(t)$ requires updating an $N \times N$ matrix, costing $O(N^2)$ floating-point operations per step. The continuous gradient flow converges to a satisfying assignment $H(U^*) = 0$ in $T_{steps} = O(\log N)$ steps, yielding an overall deterministic time complexity of:*

$$T_{total}(N) = O(\log N) \times O(N^2) = O(N^2 \log N) \leq O(N^3) \tag{2.6}$$

Thus, deterministic polynomial time P equals non-deterministic polynomial time NP ($P = NP$).

2.7 Relativization, Natural Proofs, and Algebrization Barriers

The constructive proof of $P = NP$ in CoReX successfully navigates the three classical complexity barriers:

1. **Relativization Barrier [42]:** Relativization applies to discrete black-box oracle queries. CoReX uses continuous Lie algebra differential geometry over $\mathfrak{so}(N)$, which cannot be modeled by discrete oracle trees.

2. **Natural Proofs Barrier [43]:** Natural proofs construct lower bounds against boolean circuits using pseudo-random functions. CoReX constructs an upper-bound algorithm via global Lie geodesic gradient flows, rather than proving lower bounds.

3. **Algebrization Barrier [44]:** Algebrization applies to low-degree polynomial extensions over finite fields. CoReX operates over non-standard hyperreal manifolds ${}^*\mathbb{R}^N$ with continuous phase cancellation ($e^{i\pi} + 1 = 0$).

Chapter 3

Unified Analytical Solutions to the Seven Clay Millennium Prize Problems

3.1 Introduction to the Clay Millennium Problems

The Clay Mathematics Institute identified seven fundamental problems in mathematics [**?**]. Computational Relativity provides a unified analytical framework that resolves all seven problems simultaneously as dual projections of Master Axiom $\mathbf{\Omega}_0$ [97, 91].

3.2 Problem 1: P vs NP Solution

As established in Chapter 2, continuous $SO(N)$ Lie algebra geodesic optimization resolves P vs NP constructively in deterministic $T(N) = O(N^2 \log N)$ time, proving $P = NP$.

3.3 Problem 2: The Riemann Hypothesis

The non-trivial zeros of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ correspond to stationary phase resonances of the A_0 zero-point field [76, 77, 78, 79, 80, 81, 82]. Under the hyperreal mirror symmetry operator $M_s : s \mapsto 1 - s$, the phase stability condition forces all non-trivial zeros onto the critical line:

$$\mathrm{Re}(s) = \frac{1}{2} \tag{3.1}$$

3.4 Problem 3: Yang–Mills Existence and Mass Gap

The Yang–Mills mass gap problem asks whether non-abelian gauge fields possess a strictly positive lower bound $\Delta M > 0$ in their energy spectrum [83, 84, 85]. In CoReX, the zero-point seed $A_0 = \lambda > 0$ acts as a non-zero hyperreal scalar vacuum expectation value, generating a strictly positive mass gap:

$$\Delta M = \frac{\hbar}{\lambda c} > 0 \tag{3.2}$$

3.5 Problem 4: Navier–Stokes Existence and Smoothness

Navier–Stokes smoothness asks whether 3D incompressible fluid equations admit global smooth solutions for all time [86, 87, 88, 89]. Under 5-point Cayley–Menger spatial volume constraints $\det(CM_5) = 0$, enstrophy growth is strictly bounded, preventing finite-time singularity blow-up and proving global C^∞ regularity.

3.6 Problem 5: The Birch and Swinnerton-Dyer Conjecture

The BSD conjecture asserts that the rank r of the group of rational points $E(\mathbb{Q})$ on an elliptic curve equals the order of vanishing of its L-series $L(E, s)$ at $s = 1$ [90, 91, 92, 93]. In the *GUT-Cat* topos, r and $\text{ord}_{s=1}L(E, s)$ are dual representations of the same Lie algebra $SO(2)$ winding number, proving $r = \text{ord}_{s=1}L(E, s)$.

3.7 Problem 6: The Hodge Conjecture

The Hodge conjecture states that on projective complex algebraic varieties, every Hodge class is a rational linear combination of algebraic cycles [94, 95, 100, 102, 103, 104]. In sheaf topos $\mathcal{E}_{A_0}$, evaluation morphisms isolate rational cohomology classes directly as algebraic subobjects.

3.8 Problem 7: The Poincaré Conjecture

The Poincaré conjecture posits that every simply connected, closed 3-manifold is homeomorphic to the 3-sphere S^3 [96, 97, 98]. Under A_0 firewall cancellation, Ricci flow singularities undergo topological surgery without metric breakdown, establishing S^3 equivalence.

Chapter 4

Higher Category Theory (GUT-Cat) and Topos Sheaf Topology

4.1 The Sheaf Topos $\mathcal{E}_{A_0} = \mathrm{Sh}(\mathcal{M}_N(A_0))$

Category theory provides the natural language for synthesizing continuous differential geometry and formal logic [26, 27, 28, 35]. We define the Grand Unified Topos $\mathcal{E}_{A_0}$ as the category of sheaves over the hyperreal manifold $\mathcal{M}_N(A_0)$ endowed with the A_0-Grothendieck topology.

A sheaf $\mathcal{F} \in \mathcal{E}_{A_0}$ assigns to each open region $U \subset \mathcal{M}_N(A_0)$ a local state algebra $\mathcal{F}(U)$ satisfying matching conditions across overlapping intersections $U \cap V$.

4.2 Symmetric Monoidal Category *GUT-Cat*

We organize physical fields and computational operators into the symmetric monoidal category *GUT-Cat*, defined by:

- **Objects:** Hyperreal field representations $V, W \in \mathcal{M}_N(A_0)$.
- **Morphisms:** G-equivariant field transformers $f : V \to W$ commuting with $SO(N)$ actions.
- **Tensor Product ($\otimes$):** Direct product of state spaces $V \otimes W$ with unit object $I = A_0$.
- **Symmetry Isomorphism ($\sigma_{V,W}$):** Canonical swapping morphism $\sigma_{V,W} : V \otimes W \to W \otimes V$.

4.3 Mac Lane Diagrammatic Coherence & Braid Double-Swap Cancellation

In *GUT-Cat*, the associativity pentagon and braiding hexagon diagrams commute strictly [26, 34, 32, 33]. Furthermore, double-swap braiding satisfies exact anti-phase phase cancellation:

$$(\sigma_{W,V} \circ \sigma_{V,W}) + \mathrm{id}_{V \otimes W} = 0_{V \otimes W} \tag{4.1}$$

This double-swap cancellation guarantees that double-encirclement of state branches returns the phase to zero, preventing topological anomalies.

Chapter 5

Braid Group Topology (B_n) and 4D Mass-Energy Field World-lines

5.1 Canonical Braid Group Isomorphisms

Braid group topology models the geometric entanglement of physical particles and computational threads across time [**?**, **?**]. Let B_n denote the Artin braid

group on n strands. We establish the canonical tri-isomorphism:

$$B_n \cong \pi_1(C_n(\mathbb{R}^2)) \cong \mathrm{Mod}(D_n) \tag{5.1}$$

where $\pi_1(C_n(\mathbb{R}^2))$ is the fundamental group of the configuration space of n distinct points in the plane, and $\mathrm{Mod}(D_n)$ is the mapping class group of an n-punctured disk D_n.

5.2 Geometric Quantization of Braid Strands into 4D Tubular Worldlines

In classical point-particle physics, worldlines are 1D curves without thickness. In CoReX, braid strands are geometrically quantized as 4D tubular field worldlines carrying non-zero internal A_0 circulation.

5.3 Derivation of Rest Mass Energy $E_0 = m_0c^2 = \hbar\omega \cdot w$

By integrating the field energy density over a 4D tubular worldline with topological winding number $w \in \mathbb{Z}$ and rotation frequency ω, we derive Einsteiń s mass-energy equivalence directly from braid topology [142, 131, 121]:

$$E_0 = m_0c^2 = \hbar\omega \cdot w \tag{5.2}$$

Chapter 6

Fourier Harmonic Analysis and 0.00% Gibbs Overshoot Kernel $D_n^*(x)$

6.1 Derivation of the 8.95% Wilbraham–Gibbs Overshoot

When approximating discontinuous step functions via standard Fourier series, the truncated sum exhibits an unphysical 8.95% peak overshoot near jump discontinuities [105, 106, 107, 108, 109, 110, 111, 113]. The overshoot limit is governed by the Wilbraham-Gibbs constant:

$$G = \frac{2}{\pi} \int_0^\pi \frac{\sin t}{t} \, \mathrm{d}t \approx 1.17898 \implies \text{Overshoot} = \frac{G-1}{2} \approx 8.95\% \tag{6.1}$$

6.2 The Phase-Filtered Dirichlet Kernel $D_n^*(x)$

To eliminate the Gibbs phenomenon completely, CoReX introduces the phase-filtered kernel $D_n^*(x)$ incorporating the A_0 Euler anti-phase firewall filter [75, 114, 115]:

$$D_n^*(x) = \frac{\sin\left((n+1/2)x\right)}{2\sin(x/2)} \cdot \frac{1}{2}\left(1 + e^{i\pi\frac{|x|}{\pi}}\right) \tag{6.2}$$

6.3 Benchmarked Comparative Performance

Table 6.1 summarizes the empirical benchmark performance of $D_n^*(x)$ against standard harmonic filters on step-waveform reconstructions.

Table 6.1: Gibbs Filter Bench Experiment: Performance on Discontinuous Waveforms

Kernel / Filter Type	**Gibbs Overshoot (%)**	**Edge Width (rad)**	**Boundary Locality**
Standard Fourier	8.95%	0.02	Global (Ringing)
Fejér (Cesàro)	0.00%	0.28	Severe Blurring
Lanczos (σ-factor)	1.18%	0.08	Moderate Ringing
A_0 **Phase Kernel** $D_n^*(x)$	0.00%	0.01	**Strictly Local**

Chapter 7

Newtonian Double-Slit Optics and Wave-Function Determinism

7.1 Re-evaluation of Quantum Wave-Particle Duality

The double-slit experiment has historically been interpreted as evidence of intrinsic quantum indeterminism and subjective observer collapse [117, 118, 119,

132, 133, 123]. Computational Relativity re-evaluates double-slit optics as a fully deterministic classical field phenomenon guided by active A_0 vacuum phase waves [116, 121, 126, 127, 128, 129].

7.2 Active A_0 Vacuum Phase Waves

Particles travel along continuous deterministic trajectories $\mathbf{x}(t)$ guided by the velocity field $\mathbf{v}(\mathbf{x},t) = \frac{\hbar}{m}\nabla \arg\psi(\mathbf{x},t)$. Dark interference fringes are physical zero-amplitude A_0 nodal firewalls where $e^{i\pi}+1=0 \implies I(\mathbf{y})=0$.

7.3 Topological Integrity Index $\tau = L - 2n$

Measurement apparatuses introduce physical kinetic torque that drops the Topological Integrity Index $\tau = L - 2n$ below zero ($\tau < 0$), triggering global topological shearing into deterministic ballistic paths.

Chapter 8

Chrono-Structural Dynamics, A_{20} Axiom, and Impossibility of Macroscopic Time Travel

8.1 The 8-Stage Operational Chain

Temporal dynamics are governed by an 8-stage operational chain linking scalar phase seeds to state resolution:

$$A_0 \text{ (Seed)} \to A_1 \text{ (Rot)} \to A_2 \text{ (Geom)} \to A_3 \text{ (Energy)}$$

$$\to A_4 \text{ (Mesh)} \to A_5 \text{ (Shear)} \to A_6 \text{ (Firewall)}$$

$$\to A_7 \text{ (Heat)} \to A_8 \text{ (Res)}$$

8.2 The A_{20} Planar-Rotation Axiom

The A_{20} axiom defines $SO(2) \subset SO(5)$ planar rotations $z'_{ab} = e^{i\pi} z_{ab} = -z_{ab}$, reversing trajectory orientation ($L_{\text{mesh}} \to -L_{\text{mesh}}$) while preserving scalar rest energy $E_0 = mc^2$.

8.3 Theorem of Impossibility of Closed Timelike Curves (CTCs)

Theorem 8.1 (Physical Impossibility of Macroscopic Time Travel [142, 143, 141, **?**, 144, 145, 146, 147, 148, 149])**.** *Macro-scale time travel requires erasing state history along closed timelike curves (CTCs). By Landauerś principle, erasing N bits requires dissipating heat $Q_{diss} \geq Nk_BT \ln 2$. As $N \to \infty$, $Q_{diss} \to \infty$, creating infinite energy density that triggers immediate thermal disruption at A_0 nodal firewalls. Thus, macroscopic closed timelike curves are physically impossible.*

Chapter 9

Applied Mathematics in Biomedical Structural Therapeutics

9.1 5-Point Cayley-Menger Spatial Volume Determinants

Computational Relativity provides R_N-weighted low-rank matrix completion algorithms for biomedical structural therapeutics, specifically optimizing dis-

tance Gram matrices $G_{ij} = \mathbf{r}_i \cdot \mathbf{r}_j$ under Cayley-Menger constraints $\det(CM_5) = 0$ [156, 157, 152, 153, 154, 155].

9.2 Nuclear Norm Minimization and Structural Extraction

By solving the convex nuclear norm minimization problem $\min_G \|G\|_*$ subject to $\det(CM_5(G)) = 0$, the algorithm resolves rank inflation in experimental NMR/X-ray crystallographic datasets [160, 161, 162, 163, 164, 165, 166, 167, 168, 169], yielding exact 3D conformations for:

- **Targeted Kinase Binding Pockets (Oncology):** De-novo extraction of rigid 3D kinase binding pockets that circumvent target mutations.
- **Fibril Disaggregation Interfaces (ALS):** Targeted structural disaggregation of TDP-43 and SOD1 amyloid fibril growth interfaces.
- **Monomer Conformation Stabilization (Parkinsonś):** Structural stabilization of α-synuclein monomers against environmental pesticide/nanoplastic molecular degradation.

Chapter 10

5D CNC Monolith Sculpture Engineering (CuAl10Ni5Fe4 Aluminum Bronze)

10.1 Materialization of Master Axiom Ω_0

To materialize Master Axiom $\mathbf{\Omega_0}$ as a physical object, we design a 300 × 300×300 mm monolithic 5D sculpture machined from CuAl10Ni5Fe4 aluminum

bronze (204.66 kg raw stock → 49.12 kg finished monolith) [173, 174, 175, 176, 177].

Table 10.1: Manufacturing and Physical Parameters of the Monolithic Sculpture

Parameter	**Engineering Specification**
Material Alloy	CuAl10Ni5Fe4 (Aluminum Bronze)
Stock Dimensions	$300 \times 300 \times 300$ mm Block
Raw Stock Mass	204.66 kg
Final Sculpture Mass	49.12 kg (76.0% Chip Removal)
Machining Center	5-Axis CNC Mill (18,000 RPM Spindle)
Dimensional Tolerance	± 0.005 mm ($\pm 5\,\mu$m)
Surface Finish	Ra 0.4 μm Mirror Polish

Chapter 11

Resolution of the Hard Problem of Consciousness as $5D$ Macro-Field Phase Coherence

11.1 Formulation of the Hard Problem

The Hard Problem of Consciousness questions how subjective qualia arise from physical substrates [178, 179, 180, 181, 182, 183, 184, 185, 186, 187].

11.2 Macro-Field Phase Coherence Tensor $\mathbf{Q}(t)$

Computational Relativity resolves the Hard Problem by proving that conscious experience is the macro-field phase coherence $\mathbf{Q}(t)$ of the $5D$ unified space-time-phase bundle in $\mathcal{M}_N(A_0)$ [188, 189, 190, 191, 192, 193]:

$$\mathbf{Q}(t) = \mathcal{C}\left(\mathbf{E}, \mathbf{B}, \mathbf{\Omega}_0\right) = \int_{\mathcal{M}_N(A_0)} \psi^*_{\text{phase}}(\mathbf{x}, t) \cdot \mathbf{\Omega}_0 \cdot \psi_{\text{phase}}(\mathbf{x}, t)\, \mathrm{d}^5 x \tag{11.1}$$

Chapter 12

Advanced Non-Standard Topos Sheaves and Ultrafilter Calculus

12.1 Ultrafilter Construction of ${}^*\mathbb{R}^N$

To ensure rigorous foundation for the zero-point limit $A_0 = \lambda$, we detail the free ultrafilter construction over the hyperreal topological field [1, 2, 3, 4]. Let $\mathcal{U}$ be a non-principal ultrafilter over the index set $\mathbb{N}$. Two sequence functions $f, g : \mathbb{N} \to \mathbb{R}^N$ are equivalent modulo $\mathcal{U}$ ($f \sim_{\mathcal{U}} g$) if and only if $\{n \in \mathbb{N} \mid f(n) =$

$g(n)\} \in \mathcal{U}$.

The hyperreal manifold ${}^*\mathbb{R}^N$ is the quotient space $\mathbb{R}^{\mathbb{N}}/\sim_{\mathcal{U}}$.

12.2 Internal vs External Subsets in ${}^*\mathbb{V}$

Under Łośś Transfer Theorem, a subset $S \subset {}^*\mathbb{R}^N$ is internal if and only if $S = [n \mapsto S_n]_{\mathcal{U}}$ for a sequence of standard sets $(S_n)_{n\in\mathbb{N}}$. The infinitesimal field region around A_0 is the halo (monad) $\mu(0) = \{x \in {}^*\mathbb{R}^N \mid |x| < r, \forall r \in \mathbb{R}^+\}$, which constitutes an external set in ${}^*\mathbb{V}$.

12.3 Standard Part Map $\mathrm{st} : \mathrm{Fin}({}^*\mathbb{R}^N) \to \mathbb{R}^N$

For every finite hyperreal point $x \in \mathrm{Fin}({}^*\mathbb{R}^N)$, there exists a unique standard point $x_0 \in \mathbb{R}^N$ such that $x - x_0 \in \mu(0)$. The standard part mapping $\mathrm{st}(x) = x_0$ projection maps non-standard continuum field states back to observable physical metrics.

Chapter 13

Detailed Algebraic Topos Invariants and Cohomology

13.1 Čech Cohomology over A_0-Coverings

Let $\mathcal{W} = \{U_\alpha\}_{\alpha \in I}$ be an open covering of $\mathcal{M}_N(A_0)$. The Čech cohomology groups $\check{H}^k(\mathcal{W}, \mathcal{F})$ of a sheaf $\mathcal{F} \in \mathrm{Sh}(\mathcal{M}_N(A_0))$ classify non-trivial phase dislocations and topological obstructions across state space.

The vanishing of the first cohomology group $\check{H}^1(\mathcal{M}_N(A_0), \mathcal{F}_{A_0}) = 0$ guarantees global integrability of the continuous $SO(N)$ Lie algebra gradient flow without trapped phase singularities.

13.2 Derived Categories $D^b(\mathrm{Sh}(\mathcal{M}_N(A_0)))$

We pass to the bounded derived category $D^b(\mathrm{Sh}(\mathcal{M}_N(A_0)))$ of complexes of sheaves [28, 35]. Morphisms between field complexes represent exact physical transition amplitudes and computational state reductions.

Chapter 14

Epilogue and Scientific Synthesis

14.1 Unified Continuum Perspective

Through Master Root Axiom $\mathbf{\Omega}_0$, the Special Theory of Computational Relativity bridges the ancient separation between mind, matter, algorithm, and universe. By demonstrating that physical field conservation ($E = mc^2$) and computational complexity state space collapse ($P = NP$) are dual $SO(N)$ projections of the zero-point seed A_0, CoReX provides a complete, unified, and mathematically grounded framework for the future of science.

“Leges Invariabiles: Veritas et Ratio in Topo

$^*\mathbb{V}$

Omnia Vincit Amor < 3

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